Sanity Check: UKF-Based CRM Prediction Architecture

Quick framing

Before the detailed review: this is a **latent-state estimation problem**, not a "will they buy" prediction problem. That distinction matters for almost every judgment below, so I'll come back to it repeatedly.

1. What the model's purpose actually is

You've described it correctly at a high level, but let me restate it more precisely than your own wording: this is **not a churn/win-probability model**. It's a **filter that tracks three latent, subjective buyer-state variables over time using noisy, intermittent observations**, and outputs both a point estimate and an uncertainty (covariance) for those variables at each step.

2. What it's actually estimating

Not "probability the deal closes." It's estimating **the current best guess of a hidden psychological/behavioral state** (interest, comprehension, perceived usefulness) that you believe is *correlated with* deal outcome but is not itself an outcome probability. If you later want a "probability of closing," that requires a **separate model** (e.g., a classifier or logistic mapping) that takes (μ_k, Σ_k) as input features — the UKF itself does not produce that number, and you should be explicit about this in your writeup so it isn't misread as a win-probability engine.

3. Fact-check of the math

The UKF recipe you listed (sigma point generation → propagate through g → predicted mean/covariance → sigma points through h → predicted measurement → innovation covariance S_k → cross-covariance → Kalman gain → state/covariance update) is **the standard, textbook-correct UKF algorithm.** No step is mathematically wrong. The covariance update $\Sigma_k = \hat{\Sigma}_k - K_k S_k K_k^T$ is the simplified (non-Joseph) form — fine for a course project, though it can lose positive-definiteness under numerical error; worth a footnote, not a redesign.

So: **the algorithm as written is valid UKF.** The open questions are all about whether your *application* of it to this data is justified — which is where the real critique lives.

4. Is UKF logically possible here?

Yes, in principle. UKF only requires: a state vector, a transition function g(), a measurement function h(), and noise covariances Q, R. Nothing about CRM data structurally forbids this.

But here's the catch you should not skip past: **you have not specified g() or h().** The whole justification for choosing UKF over a plain linear Kalman filter is that g/h are *meaningfully nonlinear*. Your document never defines what g actually computes (how does "activity level" nonlinearly push "Lead Interest" forward?). Without that, I can't verify UKF adds value over a linear KF — and neither can you, yet. This isn't a flaw in the math, it's a gap in the model specification that needs to be closed before implementation, not after.

5. Questionable/unjustified assumptions

- **Gaussian assumption on a bounded [0,1] variable.** UKF assumes state noise is (approximately) Gaussian, which has unbounded support. Your state is clamped to [0,1]. After a correction step, μ_k can legally leave [0,1] (e.g., a strong positive innovation pushes "Lead Interest" to 1.08). This is a real mathematical inconsistency, not a cosmetic one. Standard fixes: work in logit-transformed space and transform back, or explicitly note this as an approximation and clip with justification. Right now it's unaddressed.
- **Q and R are unspecified and, worse, unspecifiable from first principles.** In robotics/physics applications, process and measurement noise come from sensor specs or known physical variance. Here, there is no physical justification for any particular Q or R — they'd have to be fit from historical data (e.g., via maximum likelihood over observed sequences). If you hand-pick them, the filter's uncertainty output is decorative, not calibrated.
- **Ratings-as-continuous-Gaussian.** If "presentation rating" or "lead interest" is actually a discrete ordinal (e.g., 1–5 scale rescaled to [0,1]), treating it as a continuous Gaussian-observed quantity is a modeling approximation borrowed from psychometrics/IRT — defensible, but you should say so explicitly rather than assume it's obviously fine.

6. Is the state/control/measurement/noise taxonomy valid?

Mostly, but two reclassifications are warranted:

- **"Stage progression" as a control input is questionable.** A control input in the KF sense is something *applied* to drive the state forward (like a steering angle). Stage progression is arguably a *consequence* of the state (or at best a coarse proxy for it), not something the sales rep applies to change interest. It probably belongs either as part of the state itself (a 4th latent/observed dimension) or as a measurement, not u_{k-1}. "Activity level" (number of touches, calls made) is a cleaner control input, since it's an action the rep actually takes.
- **"Elapsed days / interaction timing" as a measurement is a mismatch.** Elapsed time isn't evidence *about* the state — it's a parameter that should shape the transition function g() (e.g., interest decays as a function of Δt since last contact) or set the effective timestep. Folding it into z_k means you're asking the Kalman gain to "correct" the state based on how much time passed, which doesn't fit the observation-model semantics ($h(x) \approx z$). Move it into g().

Everything else — presentation rating, updated interest/understanding/usefulness as z_k — is a legitimate direct-ish observation of the state and fits the standard measurement role fine.

7. Does the predict→measure→correct cycle make sense for CRM data?

Structurally yes, with one caveat: KF/UKF assume a well-defined update cadence (fixed or explicitly time-indexed steps). CRM interactions are irregular (meetings happen whenever they happen). This is solvable — make g() a function of Δt — but it needs to be handled explicitly, not left implicit, or your "prediction" step has no defined time horizon.

8. What it can and cannot claim

Can claim: a principled, uncertainty-aware running estimate of latent buyer-state variables that gets updated as new evidence arrives, useful as an input signal for a manager triaging where to spend attention.

Cannot claim: that it predicts deal outcome, that its Q/R-derived confidence intervals are calibrated without fitting them to historical data, that the three chosen variables are causally sufficient to describe "opportunity health," or that UKF specifically (vs. a linear KF) is necessary — that's still unproven given g/h aren't specified.

9. On adding complexity

Not recommending anything be added — but flagging that the reverse might apply: until g() and h() are shown to require the unscented transform's handling of nonlinearity, a **linear Kalman filter** is the more defensible baseline, and UKF should be justified by need, not by "it's more sophisticated." That's not a new algorithm suggestion, it's a challenge to the premise that UKF is already the right complexity tier for what's specified.

10. Decision-support framing

Your framing (trajectory indicator, not outcome guarantee) is correct and is the appropriate way to describe this class of model. Just be aware this framing only holds if μ_k is never exposed directly as a "% chance to close" in the UI — that would misrepresent a latent-state estimate as an outcome probability, which it structurally is not.

Concise verdict

Category	Assessment
Purpose of the model	Track a latent, evolving buyer-state vector using noisy CRM signals for manager decision support
What it estimates	Filtered belief (mean + covariance) over Lead Interest, Product Understanding, Product Usefulness
What it does NOT guarantee	Deal outcome, win probability, causal explanation, calibrated uncertainty without fitted Q/R
Is UKF conceptually applicable?	Yes, mathematically valid framework — but its necessity over a linear KF is unproven since g()/h() aren't yet defined
Is the state/input/measurement	Mostly, with two fixes needed: reclassify "stage progression" (state/measurement, not control) and

Category	Assessment
mapping logically valid?	“elapsed time” (belongs in g(), not z_k)
Major assumptions	Gaussian approximation on a bounded [0,1] state; Q/R chosen without data-driven calibration; ordinal ratings treated as continuous
Major corrections required	(1) Handle the bounded-state/Gaussian mismatch (e.g. logit transform); (2) move elapsed time into the transition function; (3) reclassify stage progression; (4) define g()/h() explicitly before claiming UKF (vs. linear KF) is warranted
Overall verdict	Mathematically sound as a state-estimation shell, but currently underspecified as an applied model. Fix the four items above before implementation — they’re structural, not polish.

Possible Fixes: Resolving the CRM UKF Model's Structural Gaps

Part A — Solve the bounded [0,1] state problem

Option 1: Logit transformation

Define per-variable: $y = \ln(x / (1-x))$, $x = \sigma(y) = 1 / (1 + e^{-y})$.

State vector change: Instead of filtering $x_k = [I_k, U_k, F_k]^T$ in $[0,1]^3$, filter $y_k = [\text{logit}(I_k), \text{logit}(U_k), \text{logit}(F_k)]^T$ in $\mathbb{R}^3$. The UKF/KF machinery (sigma points, mean, covariance) all operate in y-space, where the unbounded-Gaussian assumption is now consistent with the actual support of the variable.

Covariance interpretation: Σ_k is now a covariance in log-odds units, not probability units. This is less intuitive to a reader ("what does a variance of 0.4 in log-odds mean?"), so for reporting/decision-support purposes it's best to transform the mean back through $\sigma(\cdot)$ and either (a) propagate an approximate confidence interval through $\sigma(\cdot)$, or (b) report $\sigma(\mu_y)$ as the point estimate and treat Σ_y as an internal-only uncertainty measure. Do not naively report $\sigma(\Sigma_y)$ as if it were a probability-space variance — that's not how nonlinear transforms of covariance work.

Effect on g() and h(): This is actually a clean win for justifying UKF specifically: if the transition dynamics are naturally expressed in logit-space, then g becomes nonlinear when mapped back to [0,1]-space almost automatically, and the unscented transform handles that nonlinearity better than a first-order (EKF-style) linearization would. $h(x) = x$ still works fine if observations are also collected/reported in [0,1] and converted to logit-space before being compared to the predicted $\hat{z}_k$.

Incorporating [0,1]-scale observations: Convert each incoming observation ("presentation rating," "updated interest") to logit-space using the same transform before feeding it into the correction step: $z_k^{(y)} = \text{logit}(z_k^{(x)})$. This is a deterministic, invertible preprocessing step, not a modeling assumption — no controversy here.

Edge case: Logit is undefined at exactly 0 or 1. If the data ever produces a hard 0 or 1, map $[0,1] \rightarrow [\varepsilon, 1-\varepsilon]$ with $\varepsilon \approx 0.01$–0.02 before taking the logit. Document this as a known, deliberate approximation.

Practicality for a college project: Yes — it's a few lines of code (one function, one inverse function), doesn't change the UKF algorithm structure at all, and is the standard, textbook-recognized way to handle bounded states in Gaussian filters. This is not exotic.

Option 2: Explicit clipping

Let the filter run in raw [0,1]-parameterized space and clip μ_k to [0,1] after each update (and optionally clip sigma points too).

Is it mathematically acceptable? It's a common engineering hack, not a principled solution. Two specific problems:

1. Clipping the mean without adjusting the covariance is inconsistent — it silently discards information. If the unconstrained update wanted to push μ to 1.15, clipping to 1.0 leaves Σ_k as if the filter believed the true value could be 1.15 ± something, which is now nonsensical.
2. It can bias the filter over repeated corrections — if the state's "true" behavior near the boundary is asymmetric (e.g., interest saturates near 1 and can't overshoot), naive clipping doesn't model that saturation, it just papers over the symptom after the fact.

What it distorts: Primarily covariance calibration near the boundaries. Point estimates end up "close enough" for reporting purposes, but the associated uncertainty becomes meaningless exactly where you'd most want it to be trustworthy (a lead near 0.95 "very interested" or 0.05 "clearly not interested").

Defensible for a project? Only if explicitly documented as a known simplification rather than presented as a rigorous treatment. If grading criteria care about mathematical soundness, clipping should not be the only line of defense.

Option 3: Other approaches (briefly)

- **Truncated/censored Gaussian filtering** — mathematically the "correct" way to handle a hard-bounded Gaussian, but implementing a proper truncated-Gaussian Bayesian update is well beyond college-project scope and isn't necessary here.
- **Beta-distribution state filtering** (e.g., via moment matching) — conceptually elegant for [0,1] variables but is not compatible with the standard UKF machinery without substantial rework. Overkill.

These are mentioned only to show they were considered — neither is warranted.

Recommendation for Part A

Use the logit transformation (Option 1). It requires no change to the UKF algorithm itself, correctly reconciles the Gaussian assumption with a bounded variable, and gives a legitimate, citable justification for nonlinearity in g()/h() rather than an ad hoc patch. Clipping is the "easier" option, but since sigma-point propagation is already being implemented, adding two conversion functions is nearly zero extra work for a real fix. This is the one case where the "more sophisticated-sounding" answer is actually also the simpler one to implement correctly.

Part B — The Q and R problem

1. What Q means here: Q_k is the covariance of *process noise* — i.e., how much the true state can shift between observations for reasons the transition model g() does not capture (a rep's off-CRM phone call, a competitor's move, an internal client champion change). It is not a free parameter to make the filter "feel" appropriately uncertain — it's an empirical statement about how much the model of state evolution is wrong on average.

2. What R means here: R_k is the covariance of *measurement noise* — how much a single observation (a presentation rating, a self-reported interest score) diverges from the true underlying state due to noise in how CRM data is collected (subjective rater bias, coarse rating scales, mood-of-the-day effects). This is separate from process noise: R doesn't represent the state changing, it represents the "sensor" (a human filling out a CRM field) being imprecise.

3. Can they be estimated from historical CRM data? Yes, in principle, but only with **repeated observations close together in time** or **multiple sequences with known/inferable ground truth**. With a single point-in-time snapshot per opportunity, it is not possible to separate "the state moved" from "the measurement was noisy" — temporal data is required.

4. Practical estimation procedure:

- **For R:** If there are any cases where two observations of the same underlying quantity occur very close in time (e.g., two people rate the same presentation, or the same lead re-reports interest within a day where the true state plausibly hasn't moved), the sample variance of those near-simultaneous measurements is a reasonable empirical estimate of R.
- **For Q:** Run the proposed g() over historical *opportunity sequences* (a longitudinal record of the same opportunity over multiple touchpoints), compute the one-step-ahead prediction residual $x_k - g(x_{k-1}, u_{k-1}, \Delta t)$ at each step (using the *observed* — not filtered — values as a stand-in for the true state), and take the sample covariance of those residuals as the Q estimate.

5. Which statistical method is appropriate: For a college project, **residual sample covariance** (as above) is the right level of rigor — simple, explainable, defensible. Full maximum-likelihood estimation of Q/R jointly (e.g., via EM/Autocovariance Least-Squares methods used in industrial Kalman filter tuning) is a legitimate "stretch goal" to mention as future work, but not something that needs to be implemented to be taken seriously.

6. Minimum data needed: Multi-touchpoint longitudinal sequences — the same opportunity observed at 3+ points in time, for at least a few dozen opportunities, ideally covering a range of outcomes (won, lost, stalled). Cross-sectional snapshots (one row per opportunity) are insufficient for estimating either Q or R properly, because both concepts are inherently about how a single opportunity changes or is repeatedly measured over time.

7. If data is insufficient (likely, for a course project): Say so explicitly rather than presenting invented numbers as empirical. A defensible fallback: **initialize Q and R using a stated, reasoned heuristic** (e.g., "measurement noise standard deviation of 0.1 on a 0–1 scale, reflecting typical rating-scale granularity of ±1 on a 5-point scale," and "process noise standard deviation of 0.05 per week of elapsed time, reflecting an assumed gradual-drift rate"), and **explicitly label the covariance outputs as illustrative/uncalibrated** rather than statistically validated. This is honest and matches the distinction between "mathematically possible" and "empirically justified" — Q/R initialization without real data means the filter is mathematically valid but *not yet empirically calibrated*, and the writeup should say that plainly rather than let the covariance numbers imply false precision.

Practical recommendation: Use heuristic, clearly-labeled initial values for Q/R (since real longitudinal data volume is unlikely to exist within a course timeframe), and describe — as a "future work" section — the residual-covariance procedure above as what a production version would do with real historical sequences. This satisfies rigor without requiring a fabricated dataset.

Part C — Resolve the role of Stage Progression

Alternative 1 (input, u_k): In KF terms, an input is something that mechanically drives the state forward — force applied to a physical system, dosage administered to a patient. Stage progression fails this test: a rep doesn't "apply" a stage change to move interest forward; stage change is a **label CRM software assigns based on where the deal already is**. Classifying it as u_k implies causal, exogenous control that doesn't exist here. **This is the weakest option.**

Alternative 2 (additional state variable, x_k = [I, U, F, S]^T): This treats stage as something with its own (latent or semi-latent) dynamics that co-evolves with I/U/F, updated via its own transition and correction equations. This is conceptually clean *if* stage genuinely behaves like a filtered quantity (noisy, evolving, correctable by new evidence) — but CRM "stage" is usually a **discrete, categorical, often manually-set field** (Prospecting → Qualified → Proposal → Negotiation → Closed), not a continuous quantity suited to sigma-point propagation. Folding a categorical variable into a continuous Gaussian state vector is awkward and would need an ordinal-to-continuous embedding to make sense, which adds complexity not clearly justified by the project's goals.

Alternative 3 (measurement, z_k): Stage progression is **observed evidence that reflects where the underlying I/U/F state actually is** — a rep or the CRM system assigns a new stage *because* interest/understanding/usefulness (and other unmodeled factors) have reached some threshold. This matches the KF measurement semantics exactly: z_k is something observed that is a function of the hidden true state, corrupted by noise (a stage might be assigned prematurely or lag reality).

Recommendation: Treat stage progression as a measurement, via a discretized/ordinal observation model (e.g., h_stage(x) maps the continuous state to an expected stage rank, and the actual stage advancement is compared against that expectation, with an appropriately larger R to reflect its coarse, manually-assigned nature). This is the most defensible reading of "something observed because the opportunity reached a particular condition." Reserve u_k for things a salesperson actually *does* (calls made, follow-ups sent, activity count) — genuine actions, not status labels.

Part D — Properly define the state-transition model g()

Proposed structure (per-variable, in logit space $y = [y_I, y_U, y_F]^T$, consistent with Part A):

$$y_k^{(I)} = y_{k-1}^{(I)} + \alpha_1 \cdot a_{k-1} - \alpha_2 \cdot \Delta t + w_k^{(I)}$$

$$y_k^{(U)} = y_{k-1}^{(U)} + \beta_1 \cdot p_{k-1} + w_k^{(U)}$$

$$y_k^{(F)} = y_{k-1}^{(F)} + \gamma_1 \cdot \mathbb{1}[\text{presentation occurred}] + w_k^{(F)}$$

Where:

- a_{k-1} = a measure of recent sales activity (e.g., normalized contact count since last step) — this is the control input u_{k-1}.
- Δt = elapsed time since last touchpoint, entering as a **decay term** on interest (interest erodes with inactivity — this is where nonlinearity is most naturally justified: decay is often better modeled as multiplicative/exponential than linear, e.g., $y_I \leftarrow y_I \cdot e^{(-\alpha_2 \Delta t)}$ rather than subtracting a constant, since a fixed linear decay can drive log-odds arbitrarily negative regardless of Δt's scale).
- p_{k-1} = a proxy for depth of engagement in the most recent interaction (e.g., presentation occurred, questions asked) — driving Understanding.
- $\gamma_1 \cdot \mathbb{1}[\text{presentation occurred}]$ = usefulness shifts mainly in discrete jumps tied to specific presentation events, consistent with the description that usefulness perception is "relatively stable but changes after a presentation" — this justifies giving F a *sparser, event-triggered* transition rather than continuous drift like I and U.

What each coefficient means / needs to be estimated: α_1, α_2, β_1, γ_1 are not universal constants — they are exactly the kind of parameters that must be fit from historical longitudinal CRM sequences (regression of observed state changes against activity/time/presentation-occurrence). Until fit, they are illustrative placeholders, and the writeup should say so explicitly rather than presenting a specific numeric value as derived.

Numerical example (illustrative coefficients only, NOT claimed to be real):

Suppose $y_{k-1}^{(I)} = 0$ (i.e., $I_{k-1} = 0.5$), $a_{k-1} = 2$ contacts, $\Delta t = 5$ days, $\alpha_1 = 0.3$, $\alpha_2 = 0.02$:

$$y_k^{(I)} = 0 + 0.3(2) - 0.02(5) = 0.6 - 0.1 = 0.5 \Rightarrow I_k = \sigma(0.5) \approx 0.62$$

This is purely a worked mechanical example to show the math flows correctly, not a claim about real sales behavior.

Where nonlinearity genuinely enters:

1. The logit/sigmoid transform itself (state-space vs. reported-space).
2. Exponential-decay-style time dependence (if that form is chosen over linear).
3. Event-triggered (indicator function) terms for F, which are non-smooth/nonlinear by construction.

If everything instead remains additive and linear in logit-space with no Δt-decay nonlinearity, the result is essentially a **linear model in transformed coordinates** — which is a legitimate, simpler choice (see Part F).

Part E — Properly define the measurement model h()

For direct self-report-style observations (updated interest, understanding, usefulness ratings), **$h(x) = x$ is a reasonable starting point** in logit-space, since these observations are ostensibly *direct* (if noisy) reports of the same underlying quantity being tracked. This is mathematically possible and, for these specific fields, plausibly empirically justified too, since the CRM field and the state dimension are the same construct by construction.

For **presentation rating**, $h(x) = x$ is *not* obviously right, because a "presentation rating" is arguably a composite/noisy function of more than one latent variable (a good presentation might move both Understanding and Usefulness perception simultaneously). A more defensible model: $z_k^{(rating)} = h_{rating}(x) = w_U \cdot y^{(U)} + w_F \cdot y^{(F)} + v_k$, a linear combination — nonlinear only in the sense that it's expressed in transformed coordinates, not because the combination itself needs to be nonlinear.

Mathematically possible vs. empirically justified: Any h() written here is "mathematically possible" in the sense that the UKF doesn't care about its form. What's **not yet justified** is the specific weighting (w_U, w_F) or whether "rating" really is a function of exactly those two dimensions and not some other unmodeled factor — that's an empirical claim requiring validation against real data (e.g., checking whether observed ratings correlate as expected with the other directly-observed I/U/F fields). A specific h() should not be presented as final without a data-side sanity check.

Part F — Linear Kalman Filter vs UKF, applied specifically to this architecture

Criterion	Linear KF	UKF
1. Mathematical suitability	Fully suitable if g()/h() end up additive/linear in logit-space (Part D's simplest form)	Suitable for any g()/h(), including the decay/event-triggered nonlinear forms discussed
2. Represent CRM transition	Handles linear-in-logit-space drift terms fine	Needed only if genuinely nonlinear terms are adopted (multiplicative decay, indicator functions)

Criterion	Linear KF	UKF
3. Nonlinear activity/time effects	Cannot represent exponential decay or threshold/event effects without linearization error	Naturally handles them via sigma-point propagation
4. Bounded [0,1] handling	Solved equally well by the logit transform in both cases — this criterion doesn't favor either filter	Same
5. Data requirements	Lower — fewer parameters (A, B matrices) to estimate	Higher — same-order parameter count, but sigma-point tuning (κ, α, β) adds minor extra complexity
6. Parameter estimation difficulty	Easier: standard linear regression can fit A, B directly from data	Harder: nonlinear least squares or manual coefficient fitting needed for g()/h()
7. Interpretability	High — A, B are directly interpretable as "influence weights"	Moderate — nonlinear terms are less immediately interpretable, though still explainable
8. Implementation complexity	Low	Moderate (sigma point generation, weight computation, unscented transform on both prediction and update)
9. Suitability for college demonstrator	High — a correctly-executed linear KF is a complete, respectable deliverable	Also appropriate if a genuine nonlinear justification can be articulated, which raises the bar for what must be defended
10. Risk of unnecessary complexity	Low	Real risk in the current spec: if the final g()/h() turn out additive-linear, UKF adds implementation and debugging overhead for zero modeling benefit
11. Evidence needed to justify UKF	—	Demonstrated (even qualitatively) nonlinear behavior in real or plausible CRM data: e.g., interest decay that's clearly non-constant-rate, or a presentation-triggered jump that a linear model visibly can't capture

Direct answer to the key question — under what characteristics of g()/h() would UKF provide meaningful value over linear KF, specifically:

UKF earns its keep here **only if** the model adopts (a) the exponential/multiplicative time-decay term for interest (rather than linear decay), and/or (b) the event-triggered indicator term for usefulness. Both are nonlinear by construction and cannot be exactly represented by Ax + Bu. If g() is instead simplified to a fully additive, linear-in-logit-space model (drop the exponential decay, drop the indicator function, use continuous proxies throughout), **there is no principled reason left to prefer UKF over a linear KF operating in logit-space** — and choosing UKF anyway would be complexity for its own sake.

Part G — Two candidate architectures

Architecture A — Conservative / Linear KF

- **State vector:** y_k = [logit(I_k), logit(U_k), logit(F_k)]^T
- **Control vector:** u_{k-1} = [normalized contact/activity count]
- **Measurement vector:** z_k = logit-transformed [updated interest, updated understanding, updated usefulness, composite presentation rating]
- **Transition:** A = identity (or near-identity with small linear decay coefficient on I), B = per-variable activity-influence weights, both fit via linear regression on historical sequences
- **Observation:** H = identity for direct self-report fields; a fixed linear-combination row for the composite rating field
- **Q:** residual covariance from one-step-ahead prediction errors on historical sequences (or heuristic if data insufficient, per Part B)
- **R:** sample variance from near-simultaneous repeated measurements (or heuristic)
- **Elapsed time:** enters linearly, e.g., subtracted as a term inside Bu or folded into A as a Δt-scaled decay coefficient
- **[0,1] treatment:** logit transform (Part A, Option 1)
- **Stage progression:** measurement (Part C)

Architecture B — Nonlinear / UKF

- **State vector:** same y_k
- **Control vector:** same u_{k-1}
- **Measurement vector:** same z_k, plus stage-progression as an ordinal measurement with its own h_stage()
- **g():** the nonlinear form from Part D (multiplicative time-decay on Interest, event-triggered jump on Usefulness)
- **h():** identity for direct fields, weighted composite for rating, ordinal mapping for stage
- **Q, R:** same estimation approach as Architecture A
- **Elapsed time:** enters nonlinearly via exponential decay term
- **[0,1] treatment:** logit transform
- **Stage progression:** measurement, via ordinal h_stage()

Recommendation

Architecture A (Linear KF in logit-space) is the more defensible choice for the current project, for a specific reason: there is not yet empirical evidence (from Part D/F) that the nonlinear terms in g() are necessary rather than assumed. Given the instruction not to add complexity without justification, and given that the logit transform already solves the bounded-state problem regardless of which filter is chosen, **UKF's only remaining selling point — genuine nonlinear dynamics — is currently a hypothesis, not a demonstrated requirement.**

If, during implementation, a linear transition model visibly underfits observed decay/jump patterns in real or synthetic CRM sequences, *that* would be the empirical trigger to upgrade to Architecture B. Until then, starting with A and treating B as a documented "next step if justified" is the more rigorous — not the lazier — choice.

Part H — Prediction vs. probability of sale

The statement **"The filtering system estimates the current latent opportunity state. It does not itself produce a probability that the opportunity will close"** is mathematically valid and precisely correct. The UKF/KF output (μ_k, Σ_k) is a belief distribution over the three chosen latent variables — it says nothing about deal outcome unless the three variables are additionally assumed to be sufficient statistics for the outcome, which is a strong, unverified causal claim, not something the filter itself asserts.

If a conversion probability is wanted later, the correct structure is a **separate downstream mapping**:

$P(\text{conversion}) = f(\mu_k, \Sigma_k, \text{other features})$

where f could be logistic regression, a simple threshold rule, or any classifier trained on historical won/lost outcomes, using the filtered state as *input features* rather than as the probability itself. This is conceptually clean (feature extraction → downstream classifier) and is the right way to keep the concerns separated — but this is out of scope for the current phase and has not been designed here.

Part I — Final decision

1. **Best definition of the model's purpose:** A decision-support state estimator that tracks latent, evolving buyer-state variables (interest, understanding, usefulness) from noisy, intermittent CRM evidence — not an outcome predictor.
2. **Recommended state vector:** $y_k = [\text{logit}(I_k), \text{logit}(U_k), \text{logit}(F_k)]^T$
3. **Recommended control/input vector:** genuine salesperson-driven activity only (contact count, follow-ups sent) — not stage, not time.
4. **Recommended measurement vector:** logit-transformed direct self-report fields (interest, understanding, usefulness), composite presentation rating, and stage progression as an ordinal observation.
5. **Recommended treatment of elapsed time:** folded into the transition model (as a decay parameter on A/g), not treated as a measurement.
6. **Recommended treatment of stage progression:** classified as a measurement (z_k), not a control input.

7. **Recommended treatment of [0,1] variables:** logit transform, filtering done in unbounded log-odds space, converted back only for reporting.
8. **Recommended method for Q/R:** residual/sample covariance from historical longitudinal sequences if available; otherwise explicit, clearly-labeled heuristic initialization — never presented as empirically derived without the data to back it.
9. **Proposed form of g():** start linear-in-logit-space (Architecture A); nonlinear decay/event terms (Architecture B) held in reserve, adopted only if empirically motivated.
10. **Proposed form of h():** identity for direct fields; a simple fixed linear combination for the composite rating field.
11. **KF vs UKF verdict: Start with the linear KF in logit-space.** UKF is mathematically valid and available as an upgrade path, but is not currently justified by demonstrated nonlinear structure in the data.
12. **What must be demonstrated experimentally:** that the chosen transition coefficients meaningfully predict real observed state changes better than a naive baseline (e.g., "state stays constant"); that Q/R, however initialized, produce calibrated-looking prediction intervals when checked against held-out data; and, if Architecture B is pursued, that the nonlinear terms actually outperform the linear version on real sequences.
13. **What claims should NOT be made:** that the model predicts deal outcome; that its covariance is a validated statistical quantity if Q/R were only heuristically set; that the three chosen variables are a causally complete description of an opportunity's health; and that UKF is superior to a linear KF for this problem in the absence of demonstrated nonlinear dynamics.